

Weekly Progress Report - Week 02

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Domain: Core Java Programming

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Week Ending: 02

I. Overview

During the second week of my Core Java internship, I concentrated on understanding Object-Oriented Programming (OOP) concepts and applying them to the Banking Information System project. I revised Java fundamentals and then focused on classes, objects, constructors, methods, encapsulation, inheritance, polymorphism, and abstraction. I practiced implementing these concepts through small programs before integrating them into the project. I also reviewed project requirements with my mentor and prepared a modular design that would make the application easier to maintain and extend.

II. Daily Work Log

Day 1: Revised Java basics and studied the principles of Object-Oriented Programming.

Day 2: Designed Customer and Account classes with suitable attributes and methods.

Day 3: Implemented constructors, getter and setter methods, and improved data encapsulation.

Day 4: Created a menu-driven console interface and tested user input validation.

Day 5: Practiced inheritance and polymorphism through sample programs and applied reusable methods.

Day 6: Reviewed the code, corrected logical errors, optimized the class structure, and documented weekly progress.

III. Achievements

- Completed the study of major Object-Oriented Programming concepts.
- Developed reusable Java classes for the Banking Information System.
- Improved coding standards using meaningful method names and comments.
- Strengthened debugging skills by resolving compilation and runtime errors.
- Practiced over 25 Java programs covering classes, objects, arrays, methods, and constructors.
- Participated in mentor discussions and incorporated suggested improvements into the project.

IV. Challenges and Solutions

The primary challenge during this week was understanding how multiple classes communicate within a single application. Initially, I faced issues while passing object references and maintaining proper program flow. I resolved these difficulties by revising OOP concepts, practicing additional programs, and debugging step by step. I also learned how modular programming improves readability and maintainability.

V. Learning Resources

To strengthen my knowledge, I referred to Oracle Java Documentation, Bro Code Java tutorials, GeeksforGeeks, JavaTPoint, HackerRank practice exercises, and mentor guidance. These resources helped me understand theoretical concepts as well as their practical implementation.

VI. Skills Gained

This week enhanced my understanding of object-oriented design, modular programming, debugging techniques, console application development, logical thinking, and software development practices. I also improved my confidence in writing structured and reusable Java programs.

VII. Next Week's Goals

In the coming week, I plan to implement account management, deposit, withdrawal, and transaction modules for the Banking Information System. I will also focus on exception handling, validation techniques, and improving the overall quality of the application.

VIII. Weekly Summary

Overall, Week 02 was productive and helped me build a strong foundation in Object-Oriented Programming. The practical implementation of Java concepts in the Banking Information System increased my confidence and prepared me for developing more advanced project modules in the following weeks. Continuous practice and mentor feedback played a significant role in improving my technical knowledge and programming skills.